Elimination of clicks

Most of the modern declicking algorithms are based on adaptive prediction. Denote by $\epsilon(t) = y(t) - \widehat{y}(t|t-1)$ the one-step-ahead prediction error and denote by

$$\widehat{\sigma}_\epsilon^2(t) = \frac{1}{M} \sum_{i=1}^{M} \epsilon^2(t-i)$$

the local estimate of the prediction error variance (M can be set to 20, for example). Finally, let $d(t)$ be the binary output of the outlier detector

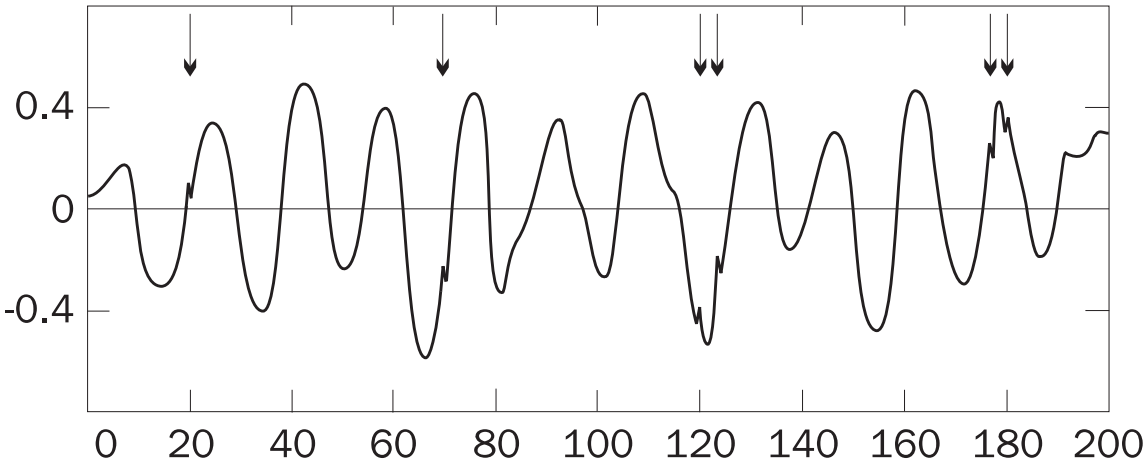
$$d(t) = \begin{cases} 0 & \text{noise impulse absent} \\ 1 & \text{noise impulse present} \end{cases}$$

The following simple decision rule can be used:

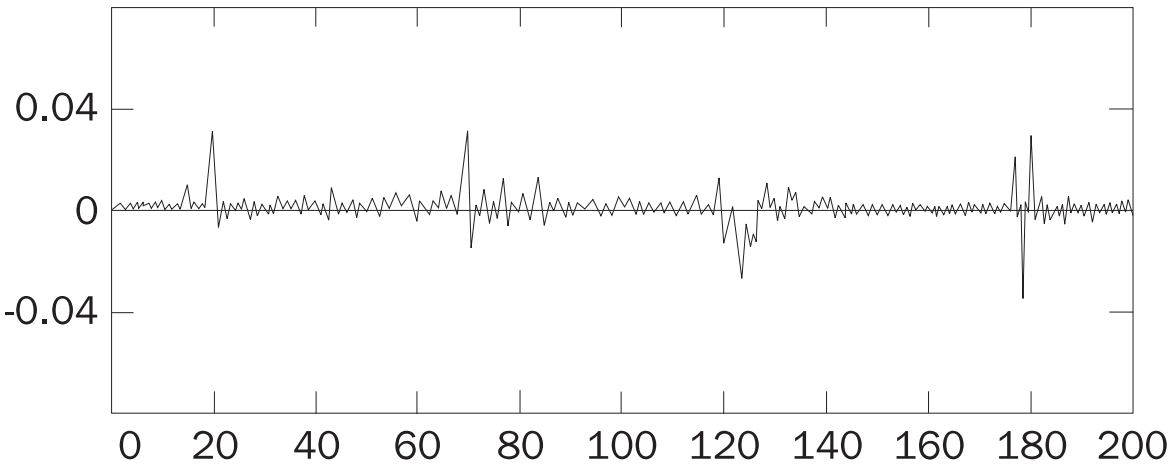
$$\widehat{d}(t) = \begin{cases} 0 & \text{if} \quad |\epsilon(t)| \leq \mu \widehat{\sigma}_\epsilon(t) \\ 1 & \text{if} \quad |\epsilon(t)| > \mu \widehat{\sigma}_\epsilon(t) \end{cases}$$

The sample is classified as an outlier and scheduled for reconstruction if the magnitude of the corresponding prediction error exceeds μ times ($\mu > 0$) its standard deviation. In audio applications the detection threshold μ has to be determined empirically; in most cases the best results are obtained for $\mu \in [3, 5]$.

Elimination of clicks



(a)



(b)

A fragment of an archived audio signal (a) and the corresponding sequence of one-step-ahead prediction errors yielded by the predictor based on the AR(10) model (b). Arrows indicate positions of audible clicks.

Reconstruction of irrevocably distorted samples

Consider an AR(r) signal $\{y(t)\}$. Suppose that one would like to reconstruct a block of m missing samples

$$y(k+1), \ldots, y(k+m)$$

based on $k \geq r$ preceding samples $\{y(1), \ldots, y(k)\}$ and $N-k-m \geq r$ succeeding samples $\{y(k+m+1), \ldots, y(N)\}$.

Reconstruction that minimizes the mean-square estimation error has the form

$$\begin{aligned}
&\widehat{y}(k+1), \ldots, \widehat{y}(k+m) \\
&\quad = \arg \min_{y(k+1),\ldots,y(k+m)} \sum_{t=r+1}^{N} \left[y(t) - \sum_{i=1}^{r} a_i y(t-i) \right]^2 \\
&\quad = \arg \min_{y(k+1),\ldots,y(k+n)} \sum_{t=k+1}^{k+r+m} \left[y(t) - \sum_{i=1}^{r} a_i y(t-i) \right]^2 \\
&\quad = J[y(k+1), \ldots, y(k+m)]
\end{aligned}$$

$$\left. \frac{\partial J[y(k+1), \ldots, y(k+m)]}{\partial y(k+i)} \right|_{\widehat{y}(k+i)} = 0, \quad i = 1, \ldots, m$$

Reconstruction of irrevocably distorted samples

Example

Suppose that the signal $\{y(t)\}$ is governed by the AR(1) model

$$y(t) = ay(t-1) + n(t)$$

and that only the sample $y(k+1)$ is missing ($m = 1$).

$$J[y(k+1)] = [y(k+1) - ay(k)]^2 + [y(k+2) - ay(k+1)]^2$$

$$\left.\frac{\partial J[y(k+1)]}{\partial y(k+1)}\right|_{\widehat{y}(k+1)} = 2[\widehat{y}(k+1) - ay(k)] - 2a[y(k+2) - a\widehat{y}(k+1)]$$

$$= 2(1+a^2)\widehat{y}(k+1) - 2a[y(k) + y(k+2)] = 0$$

$$\widehat{y}(k+1) = \frac{a[y(k) + y(k+2)]}{1+a^2}$$

Note that when linear interpolation is used, one obtains

$$\widehat{y}(k+1) = \frac{y(k) + y(k+2)}{2}$$